WALTRIP HIGH SCHOOL · ALGEBRA 1 MODEL LESSON

Factoring Trinomials

$x^2 + bx + c \quad (a = 1)$

The X-Puzzle Method

I DO model	**WE DO** together	**YOU DO** solo

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Today's Flow

Responsibility shifts from me to you — that's the whole game plan.

I DO 1

I model the thinking

You watch how I decide — every move, out loud.

Teacher does · you watch

WE DO 2

We practice together

You try each step with a partner; I guide and check.

We share the work

YOU DO 3

You prove it solo

Silent, timed, on your whiteboard. You commit — no rescue.

You own it

Responsibility shifts: me → us → you

Do Now

⏱️ 2:00

On your whiteboard — you have 2 minutes. Whiteboards up when you're done.

Multiply each pair of binomials:

1. $(x + 3)(x + 5)$

2. $(x + 6)(x - 2)$

Remember: use FOIL or the box method.

First · Outer · Inner · Last

Then combine the two middle terms into one. Keep your work — we use it in 2 minutes.

Today's Objective

Read it with me. By the end, this is exactly what you'll be able to do.

LO **Learning Objective**

I can factor a trinomial $x^2 + bx + c$ $(a = 1)$ into two binomials.

DOL **Demonstration of Learning**

Given 4 trinomials, I will correctly factor at least 3 of 4 on my exit ticket.

Essential Question

How do the b and c terms of a trinomial help us factor it?

TEKS A.10E — Factor, if possible, trinomials with real factors in the form $ax^2 + bx + c$, including perfect-square trinomials. · ELPS 2I — Demonstrate listening comprehension.

What do you notice?

Turn & Talk · 30s

Look back at your Do Now. There's a hidden pattern — find it.

$$(x+3)(x+5) = x^2 + 8x + 15$$

sum → middle: $3 + 5 = 8$

product → last: $3 \cdot 5 = 15$

$$(x+6)(x-2) = x^2 + 4x - 12$$

sum → middle: $6 + (-2) = 4$

product → last: $6 \cdot (-2) = -12$

THE BIG IDEA

The two numbers ADD to b and MULTIPLY to c. Factoring just runs that backwards.

I Do: Factor $x^2 + 7x + 12$

I DO

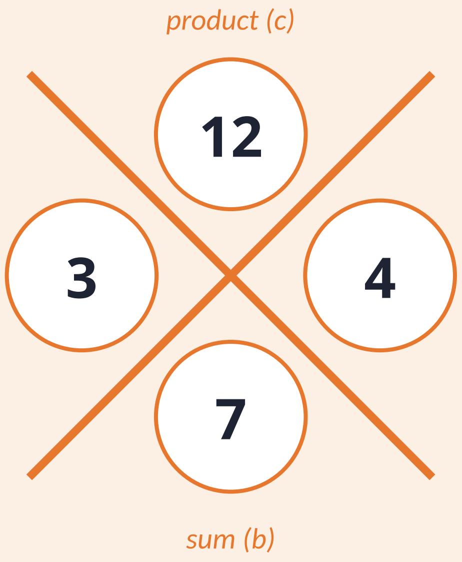
I model the thinking

Watch how I DECIDE — not just what I write.

My thinking, step by step

1. **GCF?** None — 1, 7, 12 share no common factor.
2. I need two numbers that **multiply to c = +12** and **add to b = +7.**
3. **Signs:** product +, sum + → both factors positive.
4. **Test pairs:** $1 \cdot 12 = 13$ ✗ $2 \cdot 6 = 8$ ✗ $3 \cdot 4 = 7$ ✓
5. The numbers are **3 and 4.**
6. **Write the factors:** $(x + 3)(x + 4)$
7. **Check (FOIL):** $x^2 + 4x + 3x + 12 = x^2 + 7x + 12$ ✓

The X-Puzzle



Factors:

$(x + 3)(x + 4)$

Quick Scaffold: Which signs?

Scaffolds expire ↘️

Keep this nearby for the next few problems — then we let it go.

Product (c)	Sum (b)	The two factors…	…look like
+	+	BOTH positive	(x + _)(x + _)
+	−	BOTH negative	(x − _)(x − _)
−	+	one + , one − — LARGER is +	(x + bigger)(x − smaller)
−	−	one + , one − — LARGER is −	(x − bigger)(x + smaller)

Memory hook: *Same sign? both match c's neighbor. Different signs (c is −)? the bigger number wins b's sign.*

We Do: Factor $x^2 + 9x + 20$

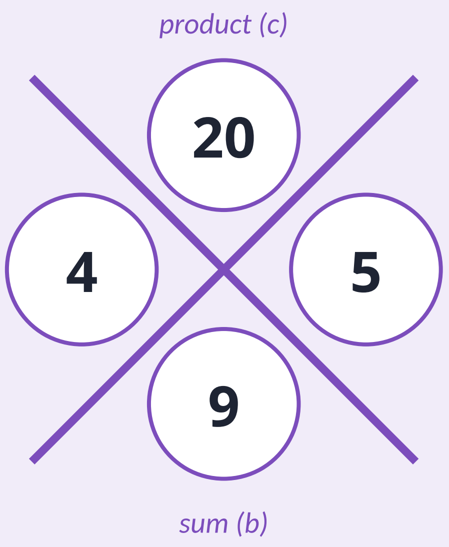
WE DO

We try it together

We'll fill the X-puzzle together. You call the numbers.

Our steps together

1. **GCF?** None.
2. **Two numbers:** multiply to +20, add to +9.
3. **Signs?** product +, sum + → **both positive.**
4. **Pairs of 20:** 1·20=21 ✗ 2·10=12 ✗ 4·5=9 ✓
5. **Numbers:** **4 and 5.**
6. **Answer:** $(x + 4)(x + 5)$



TURN & TALK · 30 sec

Why are BOTH factors positive here? Convince your partner using the sign chart.

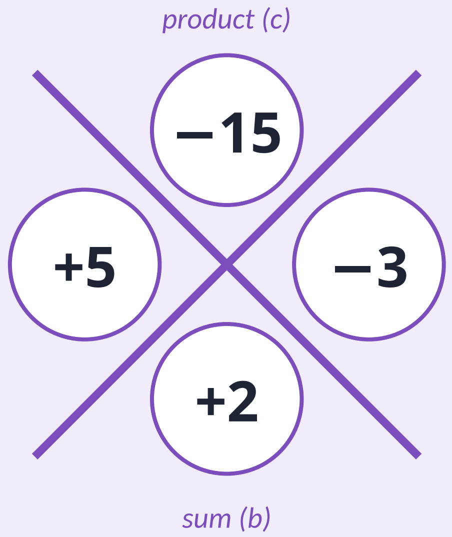
We Do: Factor $x^2 + 2x - 15$

WE DO
We try it together

Now a tougher one — the c is NEGATIVE. What changes?

Our steps together

1. **GCF?** None.
2. **Two numbers:** multiply to −15, add to +2.
3. **Signs?** c is − → **one +, one −.**
4. **Pairs of 15:** 1 & 15 (diff 14) ✗ 3 & 5 (diff 2) ✓
5. **b is + →** the bigger one (5) is +: **+5 and −3.**
6. **Answer:** $(x + 5)(x - 3)$



TURN & TALK · 30 sec

Which factor gets the + sign — and how do you KNOW? Defend it to your partner.

You Do: On your own

YOU DO

You prove it solo

Your turn to row the boat — I'm not rescuing this one.

The rules

- ✓ Silent — this is YOUR thinking.
- ✓ Show your X-puzzle for each.
- ✓ Every person commits to an answer.
- ✓ Whiteboards UP when time is called.
- ✓ Talk to your partner ONLY after — to confirm, not copy.

⏱ 3:00 · silent & timed

1. $x^2 - 7x + 12$

Factor completely. Show your X-puzzle.

2. $x^2 + 3x - 10$

Factor completely. Show your X-puzzle.

Exit Ticket — Demonstration of Learning

DOL
Exit ticket

This is what I collect. Mastery = at least 3 of 4 correct.

1. $x^2 + 8x + 15$

   factor completely

2. $x^2 - 9x + 20$

   factor completely

3. $x^2 + x - 12$

   factor completely

4. $x^2 - 4x - 21$

   factor completely

Name on top. Hand it to me on your way out. *Covers all four sign cases — that's how I know you've really got it.*

Let's stamp it.

Back to our Essential Question — you answer it now.

Essential Question

How do the b and c terms of a trinomial help us factor it?

Sentence frame: *“I factor by finding two numbers that* **multiply to c** *and* **add to b***.”*

THE STAMP

c is the PRODUCT. b is the SUM.

Find the two numbers → **that's your factor pair.** Then check the signs.

THANK YOU

Building the Minds of Tomorrow's Leaders

Ram Nation

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